



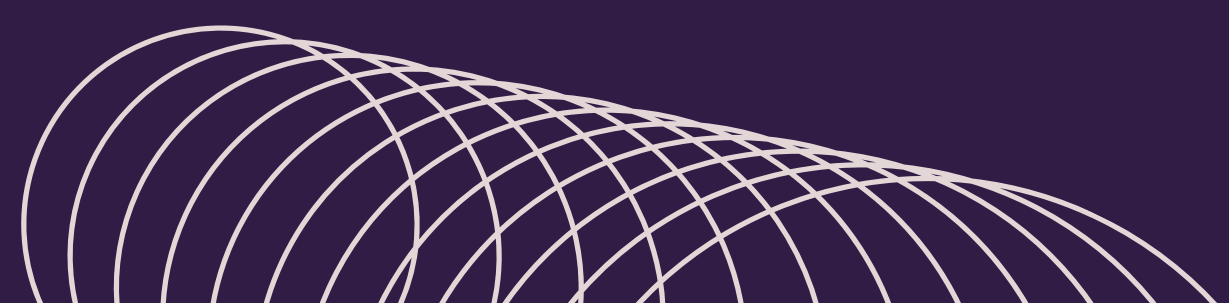
AUTOMATED CUSTOMER REVIEWS

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ABOUT DATASET

This dataset has over 34,000 reviews of Amazon products and includes 26 columns. From the 26 available columns in the dataset, we focus on the following key columns for different parts of our project



COMBINE THE DATASET

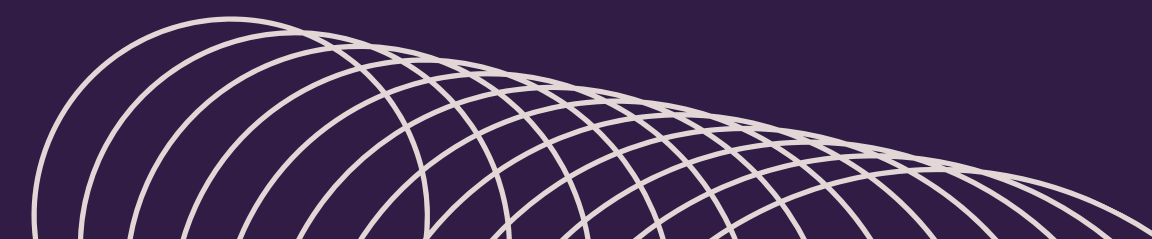
- Datafiniti_Amazon_Consumer_Reviews_of_Amazon_Products.csv
- Datafiniti_Amazon_Consumer_Reviews_of_Amazon_Products_May19.csv





SELECTED COLUMNS OVERVIEW

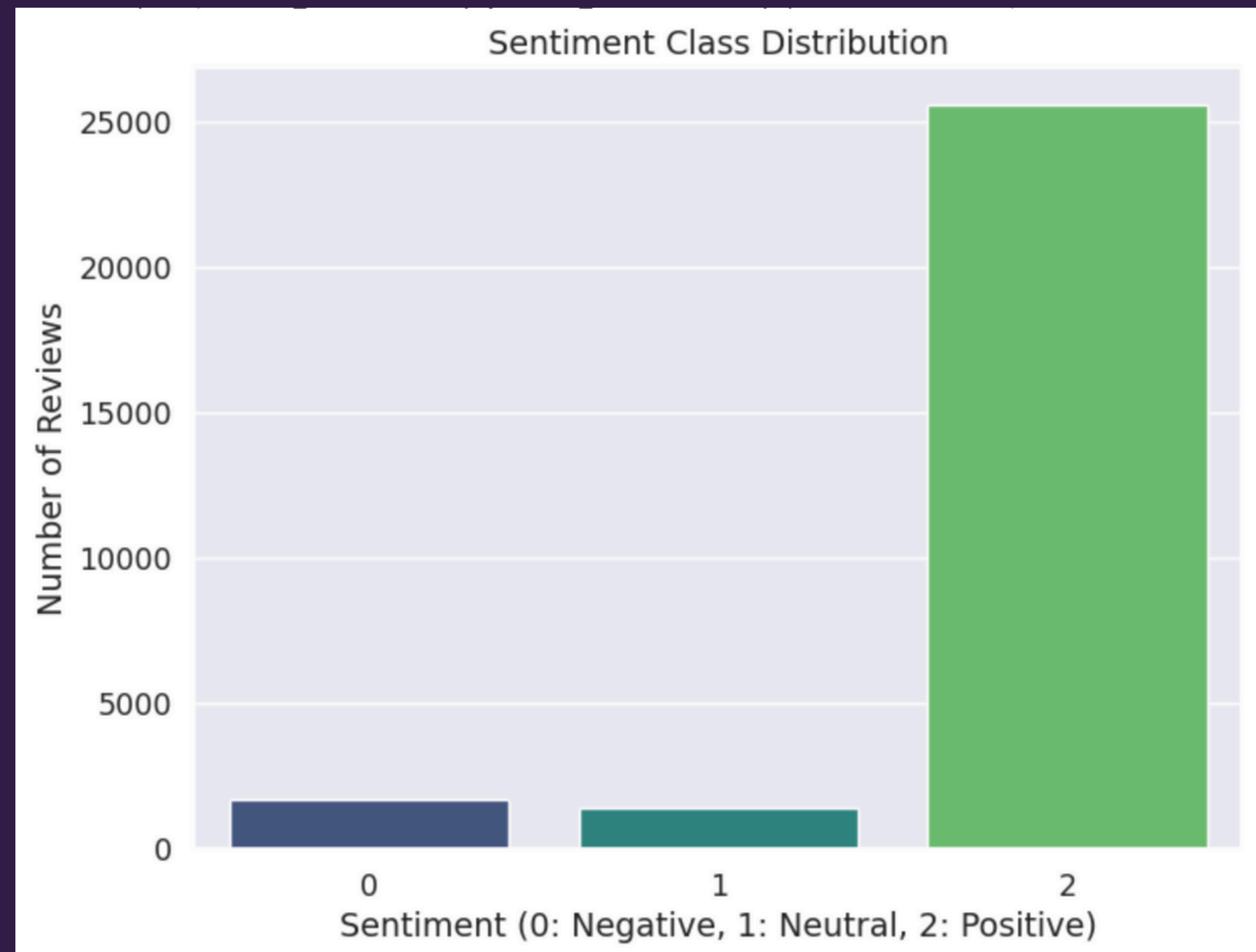


- **reviews.rating** – Used to create the label column for classification and also plays a key role in the recommender system to help identify the top 3 products.
 - **reviews.text** – Utilized in the training phase of our transfer learning model to classify reviews as positive, negative, or neutral. It's also used by the API to generate personalized recommendations for each of the top 3 products.
 - **categories** – Used to extract unique product categories for the clustering task and to generate the meta_category column.
 - **name, reviews.doRecommend, meta_category, labels** – These columns support the logic behind the recommender system, helping to identify and rank the top 3 recommended products.
- 

WHY DISTILBERT WAS A SMART CHOICE ?

- ✓ Achieved high accuracy despite noisy data → proves robustness and efficiency
- ✓ Lightweight and fast → ideal for real-world applications with limited resources
- ✓ Performs well on sentiment classification in general cases
- ✓ BERT/RoBERTa handle complex emotions better — but at a higher cost.

IMBALANCE PROBLEM



STRATEGIES TO HANDLE CLASS IMBALANCE



Sampling Techniques

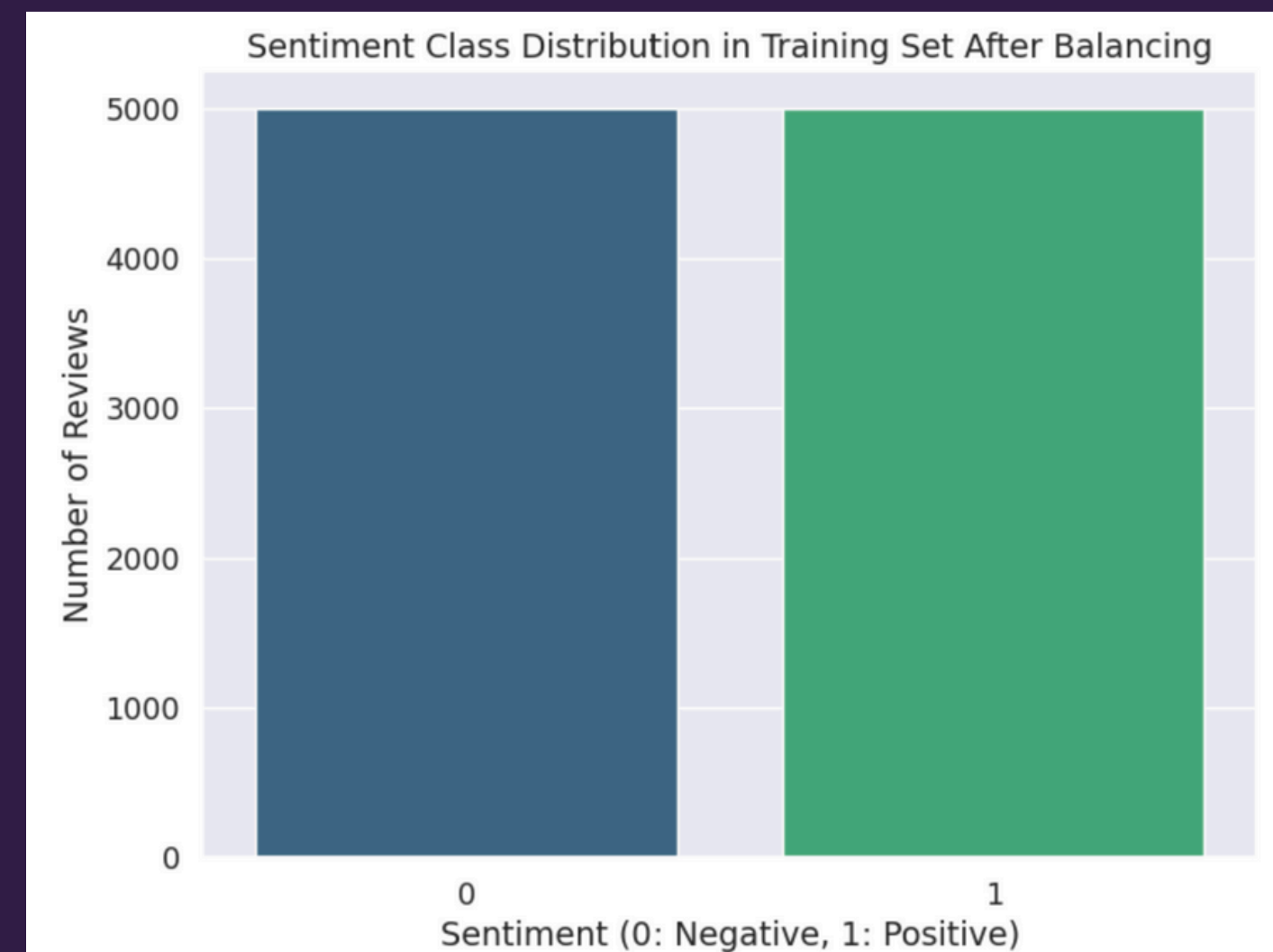
- **Oversampling:** Increase samples of minority classes
- **Downsampling:** Reduce samples from majority class



Class Weights

- **Assign higher weights to underrepresented classes in the loss function**
- **Helps the model focus on harder-to-learn categories**

SAMPLING TECHNIQUES



HYPERPARAMETER TUNING

1 *Weight Decay=0.01*

2 *Warmup Steps =250*

3 *Learning Rate = $1e-5$*

4 *Batch Size = 8*

5 *Logging Steps = 10*

6 *Epoch =7*

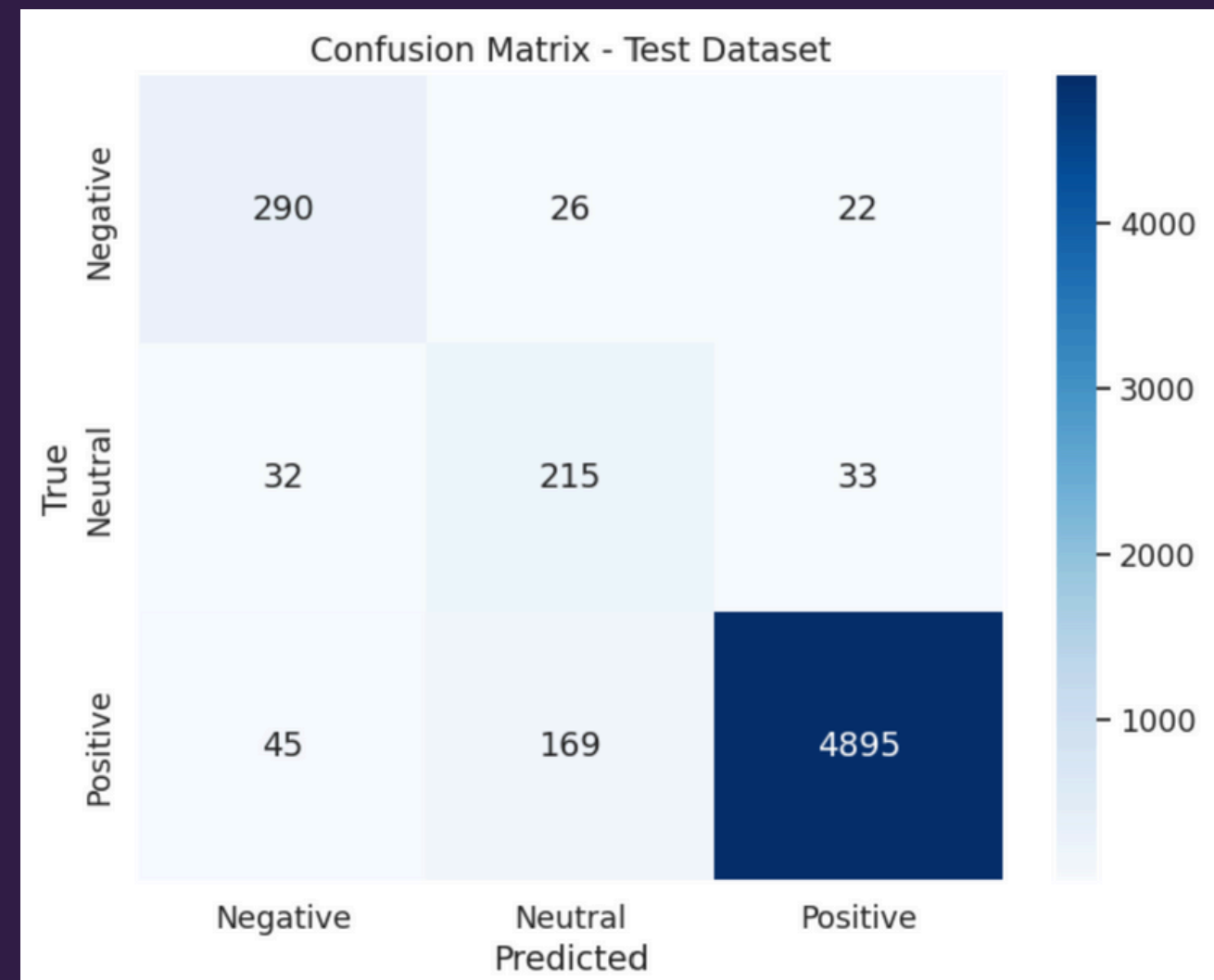
RESULT

Accuracy: 0.9429

Negative - Precision: 0.7902, Recall: 0.8580, F1: 0.8227

Neutral - Precision: 0.5244, Recall: 0.7679, F1: 0.6232

Positive - Precision: 0.9889, Recall: 0.9581, F1: 0.9733

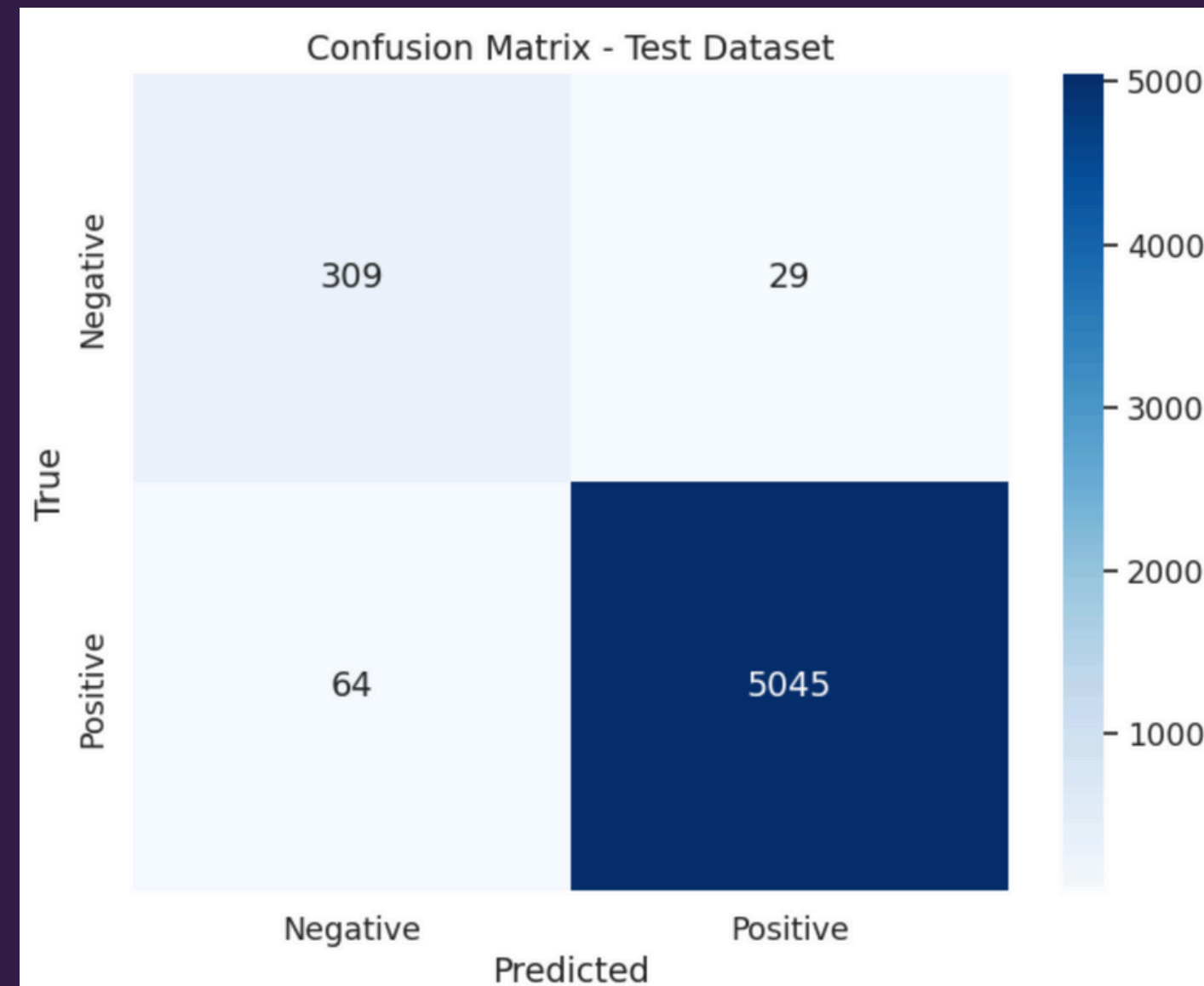


RESULT

Accuracy: 0.9829

Negative - Precision: 0.8284, Recall: 0.9142, F1: 0.8692

Positive - Precision: 0.9943, Recall: 0.9875, F1: 0.9909



CLASS WEIGHTS

- 1 *Downsample the positive class only by 8000*
- 2 *Use Class Weights approach*
- 3 *Fine tuning*

CLASS WEIGHTS

```
class_weights[0] *= 1.5
```

```
class_weights[0] *= 1.6  
class_weights[1] *= 1.8
```

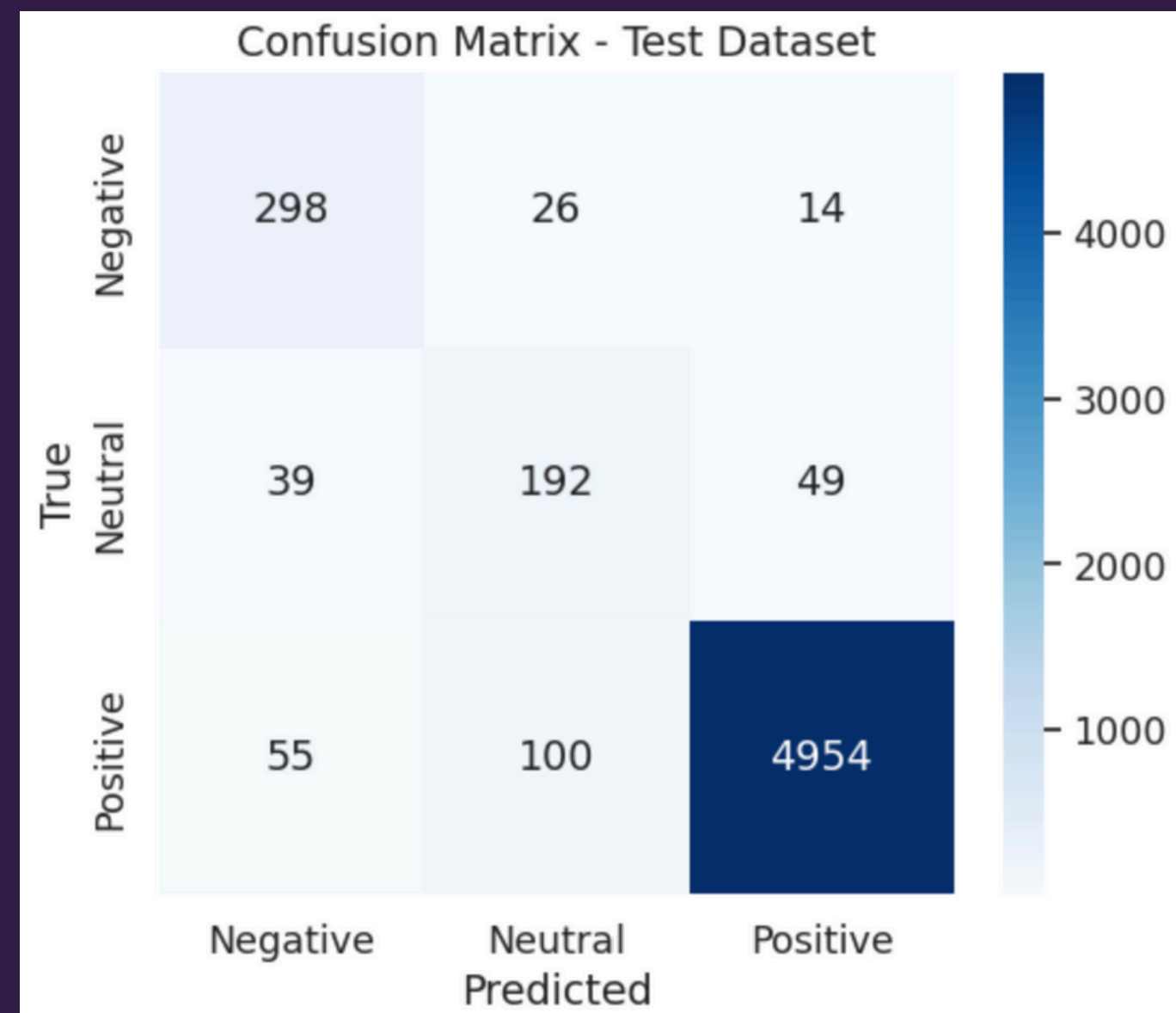

RESULT

Accuracy: 0.9506

Negative - Precision: 0.7602, Recall: 0.8817, F1: 0.8164

Neutral - Precision: 0.6038, Recall: 0.6857, F1: 0.6421

Positive - Precision: 0.9874, Recall: 0.9697, F1: 0.9785

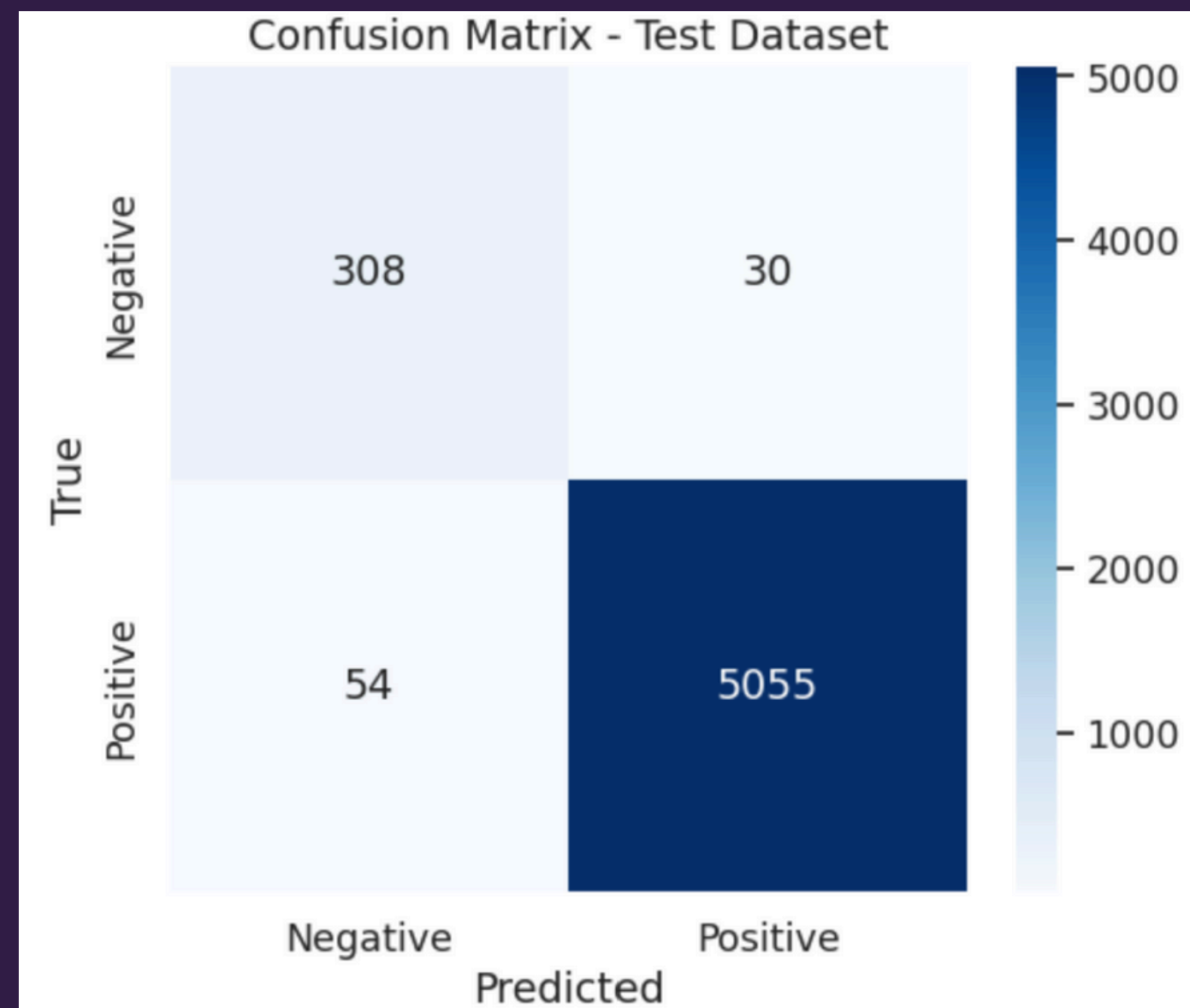


RESULT

Accuracy: 0.9846

Negative - Precision: 0.8508, Recall: 0.9112, F1: 0.8800

Positive - Precision: 0.9941, Recall: 0.9894, F1: 0.9918



CIUSTRING: HDBSCAN

Amazon Devices -	0	351	47	118	0
Home Entertainment -	351	0	56	457	17
Power & Charging Accessories -	47	56	0	12	0
Smart Home & Kitchen -	118	457	12	0	0
Storage & Office Supplies -	0	17	0	0	0

CIUSTRING: K-MEANS

Audio & Video Devices	0	1171	848	267	700	294
Electronics & Power	1171	0	729	182	1067	426
Kitchen & Storage	848	729	0	35	622	176
Office & Pet Supplies	267	182	35	0	35	11
Smart Home & Computers	700	1067	622	35	0	176
Tablets & E-Readers	294	426	176	11	176	0

REVIEW SUMMARIZATION USING GENERATIVE AI RESULT

Supplies

Avg Rating

4.6/5

Total Products

28

Positive Sentiment

4%

Market Intelligence Summary

When looking for a new tablet or e-reader, one of the top products to consider is the All-New Fire HD 8 Tablet. With a high user recommendation of 97%, this device features an 8-inch HD display, Wi-Fi connectivity, and 16 GB of storage. Despite lacking highlighted strengths, its high rating of 4.5/5 speaks for itself. The All-New Kindle Oasis E-reader also stands out, available in two versions: 32 GB and 8 GB. Both versions come with a 7-inch high-resolution display, waterproof feature, and built-in audible. While the 32 GB version is recommended by 86% users and is highlighted for being a little pricey, it has an overall rating of 4.4/5. On the other hand, the 8 GB version is highly recommended by 95% of users and boasts a slightly higher rating of 4.6/5. In the reviews, we noticed three key discussion themes. Users love their Kindles, claiming they are great, easy to use and they love using them. The Fire HD 8 Tablet is also praised for being great for kids at a good price. Lastly, many parents mentioned that they bought these devices for their kids, particularly highlighting that their daughters loved it. In conclusion, both the All-New Fire HD 8 Tablet and the All-New Kindle Oasis E-reader offer great features and are loved by users. Which one is best for you would depend on your specific needs and budget.

THANK YOU

